Time series and forecasting CHEAT SHEET

PATTERNS AND THEIR METHODS

horizontal → moving average, exponential smoothing
trend → trend line (least squares), Holt's method
seasonal → dummies, or a decomposition
trend + seasonal → dummies + t, or T × S
cyclical: waves of more than a year; hard to time
always: plot the series, name the pattern
retailer: Q4 → Q1 −24.9 %, deseasonalised +3.2 %

FORECAST ACCURACY

error $e_t = Y_t - F_t$ (+ means forecast too low)
mean error = bias: naive, 12 weeks: 5/11 ≈ 0.45

$$\text{MAE} = \frac{\Sigma|e|}{n} = \frac{61}{11} \approx 5.55$$

$$\text{MSE} = \frac{\Sigma e^2}{n} = \frac{377}{11} \approx 34.27, \quad \text{RMSE} \approx 5.85$$

$$\text{MAPE} = \frac{100}{n} \Sigma \frac{|e|}{Y} \approx 11.0\ \%$$

MAE: equal costs · MSE: big misses hurt · MAPE: compare series (not near zero) · pick the measure first
errors 0, 0, 1, 12 vs 4, 4, 4, 4: MAE 3.25 vs 4, MSE 36.25 vs 16: the measures can disagree

JUDGE ON A HOLDOUT

build on the older data, test on the latest (holdout)
weeks 1–8 fit: curve MAE 2.0 < average 2.88
weeks 9–12: average 3.25, the curve > 400
trust only errors on data the method hasn't seen

MOVING AVERAGES

$$F_{t+1} = \frac{Y_t + \ldots + Y_{t-k+1}}{k} \quad \text{(the last k periods)}$$

week 4: (51 + 46 + 53)/3 = 50
k = 3: MSE 21.22 (naive 33.67, weeks 4–12)
longer k: smoother, slower (k = 5: 16.82 vs 21.57)
weighted: weights sum to 1, newest first
3/6 × 56 + 2/6 × 52 + 1/6 × 45 ≈ 52.83

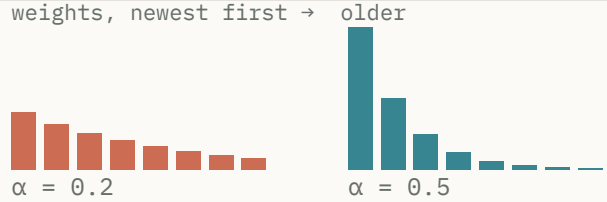
WARM-UP: ERROR AND LEAST SQUARES

errors 4, −2, 3, −5, 1: mean 0.2 (they cancel)
MAE 3 · mean square 11 · √11 ≈ 3.3

$$b_1 = \frac{\Sigma(t-\bar{t})(y-\bar{y})}{\Sigma(t-\bar{t})^2}, \quad b_0 = \bar{y} - b_1\bar{t}$$

sign-ups 3, 5, 4, 7, 8: $\hat{y} = 1.8 + 1.2t$, month 6: 9.0

EXPONENTIAL SMOOTHING



$$F_{t+1} = \alpha Y_t + (1-\alpha)F_t = F_t + \alpha\, e_t$$

α = 0.2: $F_3 = 0.2 \times 46 + 0.8 \times 51 = 50$, $F_4 = 50.6$
weights α, α(1 − α), α(1 − α)² …: 0.2, 0.16, 0.128
MSE: α 0.2 → 15.81 · α 0.5 → 20.33 (noisy: small α)
choose α on past data, check it on a holdout
it lags a trend: heat pumps, α 0.5, mean error +4.21

SMOOTHING IN A SPREADSHEET AND PYTHON

moving average: =AVERAGE(B2:B4), filled down
smoothing: =B1*B5 + (1−B1)*C5, filled down
errors: =B5−C5 · =ABS(D5) · =D5^2 · =ABS(D5)/B5
Python: s.rolling(3).mean().shift(1)
s.ewm(alpha=0.2, adjust=False).mean()

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TREND PROJECTION

$T_t = b_0 + b_1 t,\ t = 1, 2, \ldots, n$ (least squares)

heat pumps 23 ... 41: $b_1 = \frac{105}{42} = 2.5,\ b_0 = 31.25 - 2.5 \times 4.5 = 20$

year 9: $20 + 2.5 \times 9 = 42.5$; year 10: 45

=SLOPE(y, t) · =INTERCEPT(y, t)

=FORECAST.LINEAR(9, y, t) · =TREND(y, t, new t)

curved: steady % growth → line through log y

HOLT'S LINEAR TREND METHOD

$L_t = \alpha Y_t + (1 - \alpha)(L_{t-1} + b_{t-1})$

$b_t = \beta(L_t - L_{t-1}) + (1 - \beta) b_{t-1}$

$F_{t+k} = L_t + k\, b_t$; start $L_1 = Y_1$, $b_1 = Y_2 - Y_1$

α 0.5, β 0.3, year 4: F 29, Y 30 → L 29.5, b 2.15

year 9: 42.58, year 10: 45.15 (line: 42.5, 45)

SEASONAL DUMMY VARIABLES

k seasons → k − 1 dummies (one is the baseline)

no trend: $\hat{Y} = 50 - 8\ \text{Qtr1} + 16\ \text{Qtr2} + 40\ \text{Qtr3}$

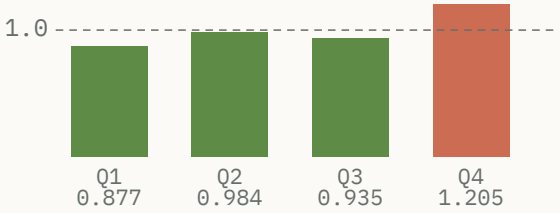
each b = that quarter's mean − Q4's (42, 66, 90, 50)

with trend: $\hat{Y} = 25.45 - 7.845\ \text{Qtr1} - 5.1\ \text{Qtr2} - 6.425\ \text{Qtr3} + 0.375t$ (=LINEST: reverse order)

fixed amounts: swings that grow → multiplicative

monthly: 11 dummies + t, 13 coefficients

DECOMPOSITION IN FIVE STEPS



$Y_t = T_t \times S_t \times I_t$ (additive: T + S + I)

1 centred MA: (21.875 + 22.25)/2 = 22.0625

2 ratio Y/CMA: 20.6/22.0625 ≈ 0.934

3 average by quarter, scale to mean 1:
0.877, 0.984, 0.935, 1.205 (=AVERAGEIF)

4 deseasonalise Y/S, fit T = 20.51 + 0.383t

5 forecast T × S: 27.0, 26.0, 34.0 ($M)

AN HONEST RANGE

fit MAPE 1.7 % · holdout MAPE 4.75 %: ≈ 3 ×

80 % range: F × (1 ± 1.28 × holdout RMS % error)

RMS ≈ 5 % → ± 6.4 %; wider further ahead

cycles: CMA/trend; needs many years of data

CASE: TALLOWFIELD SOFTWARE (FICTIONAL)

48 months of tickets; holdout: year 4

holdout MAPE: smoothing 8.9 %, line 8.2 %,
last year's month 9.1 %, decomposition 3.1 %

refit on 4 years: Jan ≈ 25,800, Jun ≈ 18,600; ±5 %

agents = ROUNDUP(upper/800): 34 in Jan, 25 in Jun

25 permanent + 50 temporary agent-months (408 flat)

SPREADSHEET AND PYTHON

AVERAGE · AVERAGEIF · ABS · SUM · SQRT · ROUNDUP

SLOPE · INTERCEPT · FORECAST.LINEAR · TREND · LINEST

NORM.S.INV(0.9) = 1.2816 (an 80 % range)

Python: numpy.polyfit(t, y, 1) · numpy.linalg.lstsq

statsmodels seasonal_decompose (multiplicative)